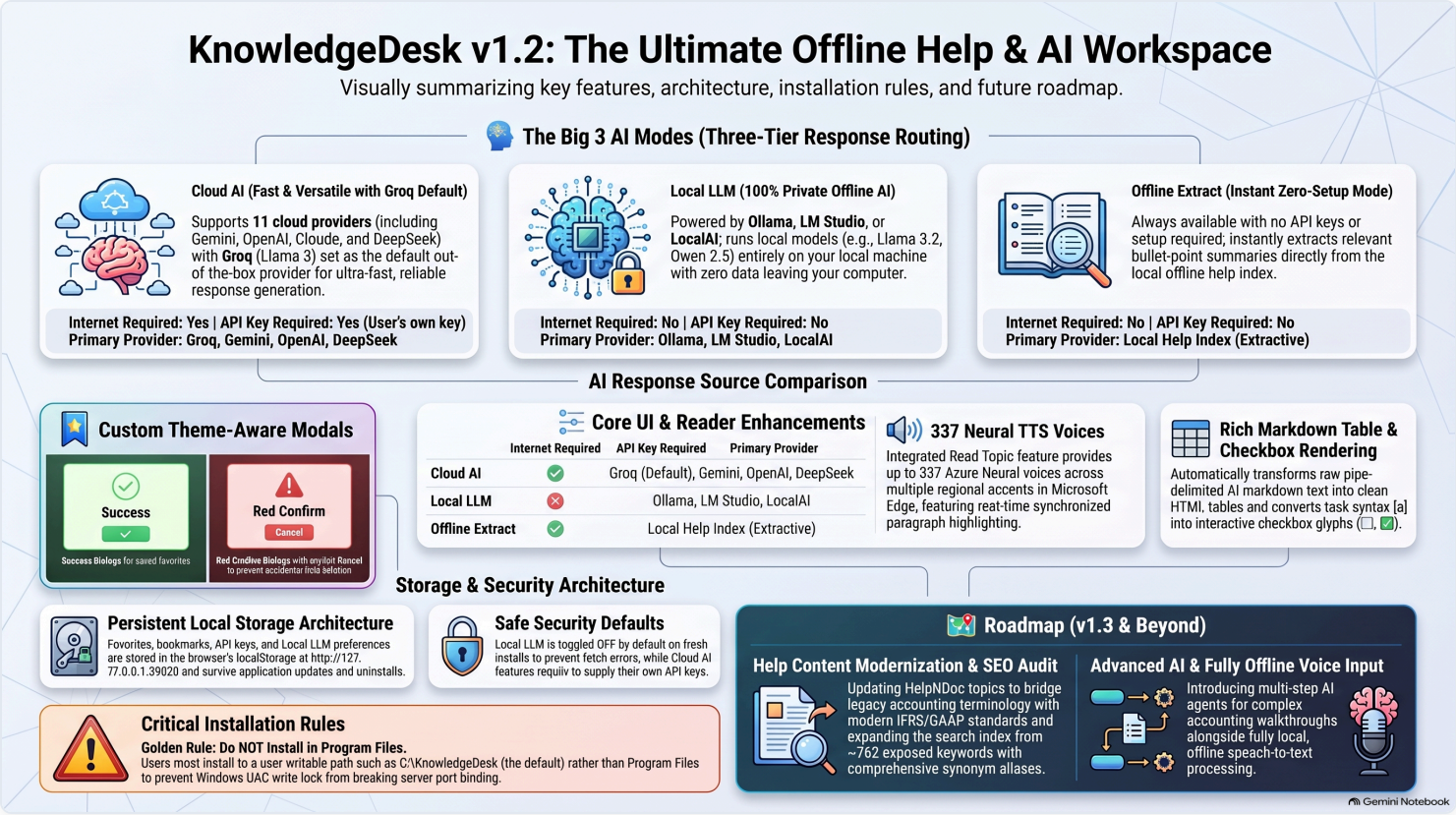
KnowledgeDesk — Offline Help with Cloud AI & Local LLM — User Guide

TurboCASH / osFinancials KnowledgeDesk — Offline Help with Cloud AI & Local LLM • **Version 1.2** • *Interface Tour & Feature Reference*

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1. Welcome



KnowledgeDesk v1.2 at a Glance. *A visual overview of the AI modes, UI features, installation rules, and upcoming roadmap.*

📚 Additional Learning Resources

Want to test your knowledge or share these guides with your team? We have prepared interactive study aids and external resources.

- 🎥 **Video Overview:** Watch the KnowledgeDesk v1.2 Feature Tour
- 🎧 **Audio Overview (Podcast):** Listen to the KnowledgeDesk v1.2 Deep Dive
- 🧠 **Interactive Quizzes** Test your knowledge on Google NotebookLM
  🧠 **Interactive Flash Cards** Test your knowledge with Flash Cards on Google NotebookLM
- 🧠 **Interactive Frequently Asked Questions (FAQ):** Frequently Asked Question on Google NotebookLM

> 💡 **A Note on Accounts:** KnowledgeDesk is a zero-login, offline-first application. You do not need an account to browse, search, or use the Local LLM.
>
> However, accessing the **Interactive Quizzes/Flashcards** (via NotebookLM) or using **Cloud AI** (via Gemini/Groq) requires you to sign in with a free Google or provider account.

TurboCASH / osFinancials KnowledgeDesk — Offline Help with Cloud AI & Local LLM is a modern, open-source documentation workspace designed to bridge the gap between traditional offline user manuals and next-generation AI assistance.

Built to run seamlessly in your browser, KnowledgeDesk transforms static topic pages into an interactive reference desk—combining fast offline content navigation with in-page topic favoriting, one-click topic printing, multi-dialect Text-to-Speech audio playback, dynamic themes, real-time search, and targeted AI.

What you can do

- **Browse & Search:** Navigate the complete offline documentation using the Contents tree or the Advanced Search engine.
- **Listen:** Use the 🔊 **Read Topic** engine, which supports up to 327 distinct voice options (in Microsoft Edge) covering regional accents, dialects, and male/female tones.
- **Favorites:** Star important topics with one click to save them to your ⭐ **Favorites** tab for instant access later.
- **Print:** Generate a clean, unformatted paper document of the *currently selected topic* using 🖨️ **Print Topic**.
- **Summarize:** Generate an instant offline summary from the local documentation, or upgrade to an AI-generated summary using Cloud AI or a Local LLM.
- **Chat:** Ask context-aware questions about accounting concepts or TurboCASH/osFinancials features in the 🤖 **AI Assistant**.
- **Offline AI:** Run AI completely offline using a local model (Ollama) — no cloud, no API key required.
- **Copy & Paste:** Copy any summary — local, Cloud AI, or Local LLM — into your favorite app.
- **Customize:** Toggle between crisp **Light Mode** and custom **Dark Mode** without ever reloading the page.
- **Discover:** Hover your mouse over any button, field, or icon to see a helpful tooltip explaining exactly what it does.

Two Modes of AI (Optional Upgrades)

The **Summarize Topic** and **AI Assistant** features will always provide an instant offline summary based on the local help documentation. You may optionally activate the following **two modes of AI** (by entering API Provider Keys or setting up a Local LLM) to unlock more capable, conversational answers:

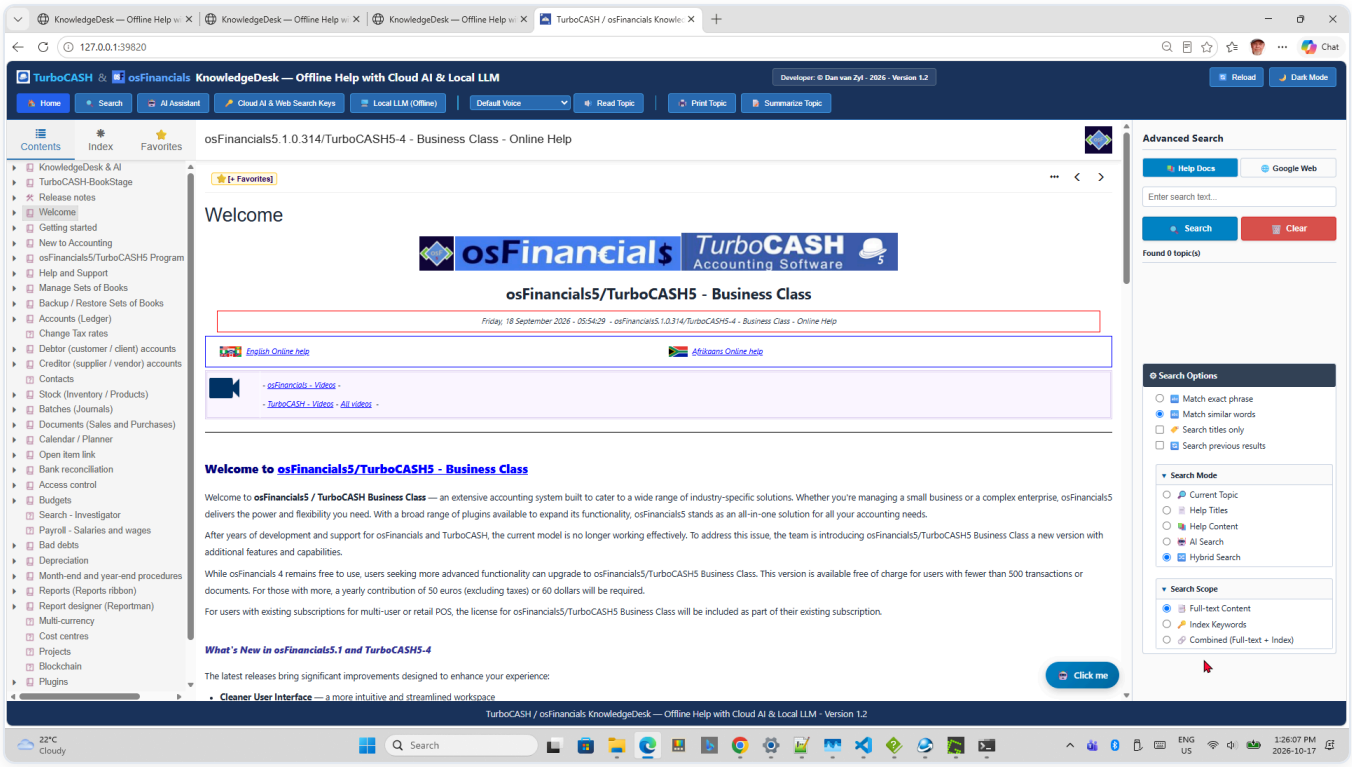
MODE	REQUIRES	PRIVACY	SPEED
☁️ **Cloud AI**	API key from a cloud provider (Groq, Gemini, OpenAI, DeepSeek, etc.)	Data sent to provider	2–5 seconds
🖥️ **Local LLM**	Ollama installed (free)	100% on your machine	5–20 seconds (warm)
📄 **Offline Extract** (Default)	Nothing	100% on your machine	Instant

> 💡 **New to the app?** Everything works without any configuration. You can browse, print, listen, and search the entire manual offline. Cloud AI and Local LLM are optional enhancements that unlock more capable, conversational responses.

2. The Main Interface

When you first launch KnowledgeDesk (or refresh the browser), you will briefly see a **Splash Screen** with a loading spinner and the TurboCASH / osFinancials branding.

This is the app connecting to your browser's Local Storage to load your saved API keys, Favorites, and theme preferences. It only takes a moment, and the screen will automatically fade away once your workspace is ready.



The main KnowledgeDesk interface at first launch

The application window is divided into five distinct areas to help you find what you need quickly:

1. **Header Bar (Top)**
   Displays the app branding, version number (v1.2), and global controls:
   - **Branding Links:** The **TurboCASH** and **osFinancials** text logos are clickable. Clicking either one will open their respective official website in a new browser tab.
   - 🔄 **Reload:** Reloads the entire KnowledgeDesk application. This refreshes the Contents and Index (table of contents) and returns you to the Welcome topic.
   - 🌙 **Dark Mode / 🌞 Light Mode:** Toggles the interface theme.

2. **Navigation Bar (Below Header)**
   Your main toolbox. From left to right:
   - 🏠 **Home:** Returns to the Welcome screen.
   - 🔍 **Search:** Opens the Advanced Search panel.
   - 🤖 **AI Assistant:** Opens the chat interface.
   - 🔑 **Cloud AI & Web Search Keys:** Where you configure your API keys.
   - 🖥️ **Local LLM (Offline):** Settings for local offline AI.
   - **Default Voice:** Choose a voice for the text-to-speech reader.
   - 🔊 **Read Topic:** Reads the current help page aloud.
   - 🖨️ **Print Topic:** Prints the page or saves it as a PDF.
   - 📝 **Summarize Topic:** Generates a summary of the currently selected topic. By default, it creates an offline summary from the local help content. If you have saved an API key or enabled a Local LLM, it will automatically upgrade to an AI-generated summary.
3. **Left Pane (Navigation)**
   Browse the manual using three tabs:
   - 📚 **Contents:** A tree view of all help chapters.
   - 🔡 **Index:** An alphabetical list of keywords.
   - ⭐ **Favorites:** Your bookmarked pages.
4. **Center Pane (Reading Area)**
   The main workspace where the help documentation is displayed. Links inside topics will open right here.
5. **Right Pane (Contextual Panel)**
   This area changes based on what you click in the Navigation Bar. By default, it shows the **Advanced Search** panel. You can also use the tabs at the top of this pane to switch between **Help Docs** and **Google Web** search.

> 💡 **The "Click Me" Button:** You will notice a floating blue **Click me** button in the bottom-right corner of the screen. This is a quick shortcut to open the AI Assistant or perform a web search without using the main navigation bar.

The interface is divided into four main areas:

2.1 — Header Bar

The top of the window shows:

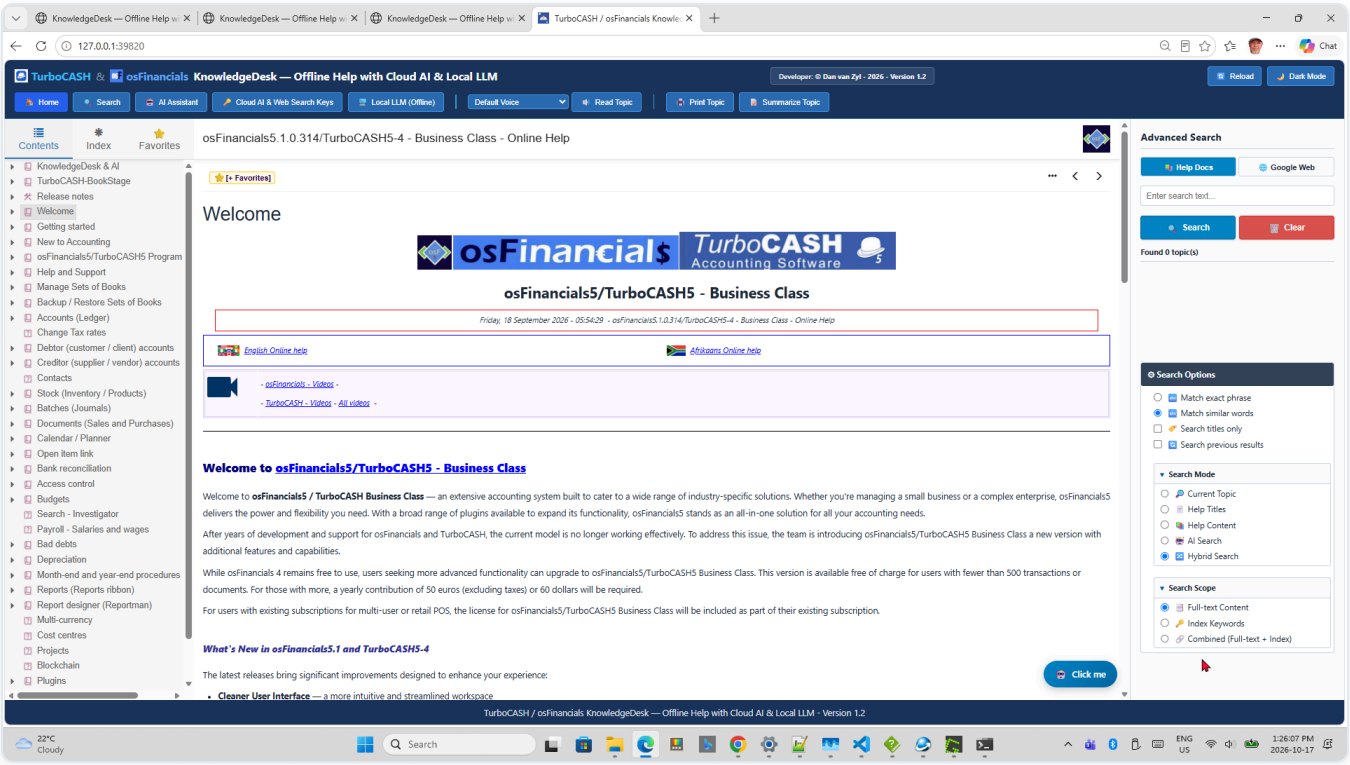
- **Branding** — TurboCASH, osFinancials, and the KnowledgeDesk subtitle
- **Developer info** — version number (v1.2)
- 🔄 **Reload:** Reloads the application, refreshes the Contents (table of contents), and returns to the default Welcome topic.
- 🌙 **Dark Mode** / 🌞 **Light Mode** — toggle the theme (see Section 3.3)

2.2 — Navigation Bar

The button row below the header gives quick access to:

BUTTON	OPENS
🏠 Home	The Welcome help topic
🔍 Search	Advanced search panel
🤖 AI Assistant	Chat with the AI
🔑 Cloud AI & Web Search Keys	API key configuration
🖥️ Local LLM (Offline)	Offline AI settings + toggle
🔊 Read Topic	Text-to-speech (with Pause / Resume / Stop)
🖨️ Print Topic	Print or export the current topic as PDF
📝 Summarize Page	AI-generated summary of the current topic

2.3 — Left Pane (Navigation)



Left pane with tabs for Contents, Index, Search, and Favorites

The left pane offers several tabs for finding content:

- 📚 **Contents** — tree view of all help topics, organized by chapter
- 🔤 **Index** — alphabetical keyword index (see Section 4.4)
- ⭐ **Favorites** — your bookmarked topics (see Section 5)

> 💡 **About the Search tab.** The embedded help system originally shipped with a Search tab in the left pane. KnowledgeDesk replaces it with ⭐ **Favorites**, because the app provides its own Advanced Search in the right pane (opened by the 🔍 **Search** navigation button). This avoids two competing search boxes and gives bookmarks a natural home next to Contents and Index.

2.4 — Center Pane (Help System)

The main content area displays the currently selected help topic. Links inside topics navigate to related sections, and external links open in a new browser tab.

TOPIC NAVIGATION CONTROLS

In the top-right corner of the reading pane, you will see a small row of navigation icons. These allow you to navigate the manual sequentially without using the left-hand Contents tree:

- **...** (Ellipses): Opens a dropdown menu showing all the headings (H1-H4) inside the current topic, allowing you to jump directly to a specific section.
- **^** (Up Arrow): Moves you back to the main parent topic (chapter) in the Contents tree. This is useful when you are reading a sub-topic and want to quickly return to the main chapter overview.
- **<** (Left Arrow): Moves to the **previous** topic in the manual.
- **>** (Right Arrow): Moves to the **next** topic in the manual.

PANEL	PURPOSE
🔍 **Advanced Search**	Full help search with filters, AI Search, and Web Search options.
🤖 **AI Assistant**	The chat interface to ask questions about the documentation.
🔑 **Cloud AI & Web Search Keys**	Configure API keys for Gemini, OpenAI, Groq, etc.
🖥️ **Local LLM (Offline)**	Configure and enable offline AI using Ollama.

> The right pane can be resized by dragging the vertical bar between the center and right panes. Click 🏠 **Home** to hide the right pane entirely.

3. Browsing the Help

3.1 — Navigating to Topics

The main reading area (Center Pane) displays whichever topic you select. You are not limited to using the Contents tree—you can load a topic into the Center Pane using any of the following methods:

- 📚 **Contents Tab:** Click a chapter to expand it, then click a topic title to load it.
- 🖨️ **Index Tab:** Click any keyword to jump directly to its associated topic.
- ⭐ **Favorites Tab:** Click any of your saved bookmarks to reload that page.
- 🔍 **Advanced Search:** Click any result in the right-hand panel to load the matching topic.
- **Internal Links:** Click any blue link inside the text of a topic to jump to a related page.

> 💡 **Note on Keyboard Shortcuts (like F4):** You may see instructions inside the help text telling you to press a function key (e.g., "*Press F4 to open the Debtors help section*"). These are keyboard shortcuts built into the **TurboCASH / osFinancials software itself**, not KnowledgeDesk. They will only work when you are actively using the accounting software, not while browsing this Help system.

3.2 — Following links

Links inside help topics behave normally:

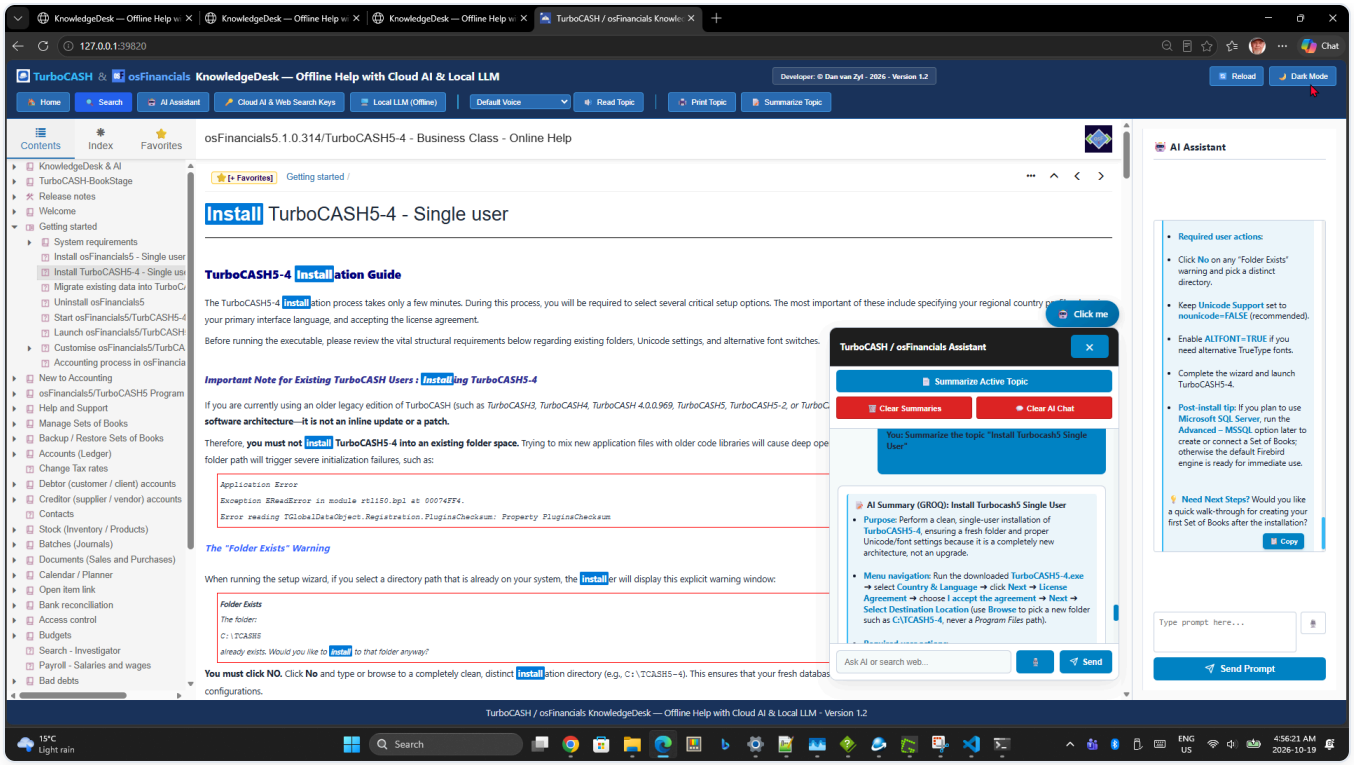
- **Internal links** — load another help topic in place
- **External links** — open in a new browser tab

3.3 — Adjusting the theme (Light / Dark Mode)

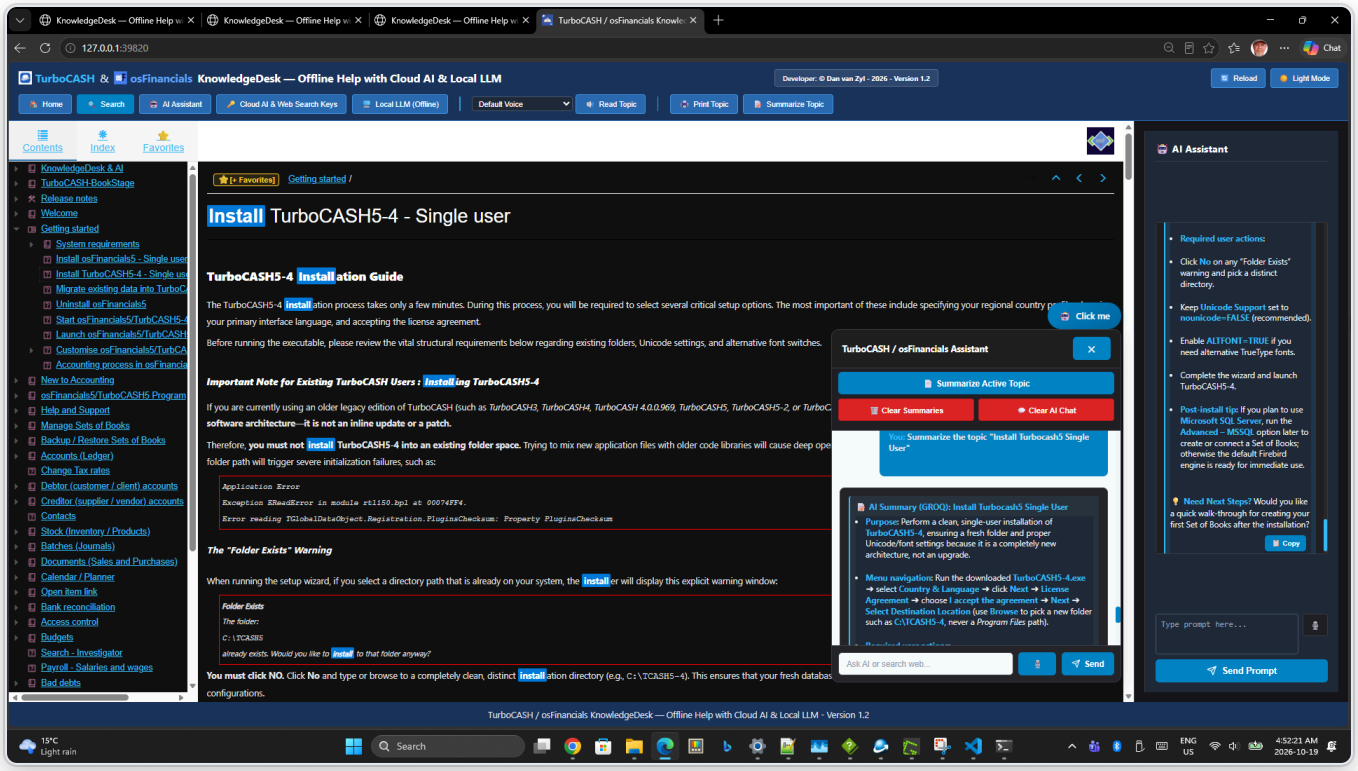
KnowledgeDesk ships with two themes: **Light Mode** and **Dark Mode**.

> 🌓 **Default Theme Behavior:**
> On a brand-new installation, KnowledgeDesk opens in **Light Mode** by default. If you prefer Dark Mode, simply click the 🌙 **Dark Mode** button in the header.
>
> **Your choice is saved automatically.** If you close the server and app while in Dark Mode, the next time you launch KnowledgeDesk, it will remember your preference and open directly in Dark Mode.

The two figures below show the same interface in each theme. Notice how every panel — header, left pane, centre pane, right pane, tables, and callouts — adapts.



Light Mode *— the header button reads* 🌙 ***Dark Mode****. Click it to switch.*



Dark Mode *— the same interface with dark panels and light text. The header button now reads* 🌞 ***Light Mode****.*

HOW TO SWITCH

1. Look for the theme button in the top-right of the header — it reads 🌙 **Dark Mode** while you're in light theme, and 🌞 **Light Mode** while you're in dark theme.
2. Click it once to flip to the other theme.

3. The setting is remembered across sessions — the next time you open the app, it restores your last choice.

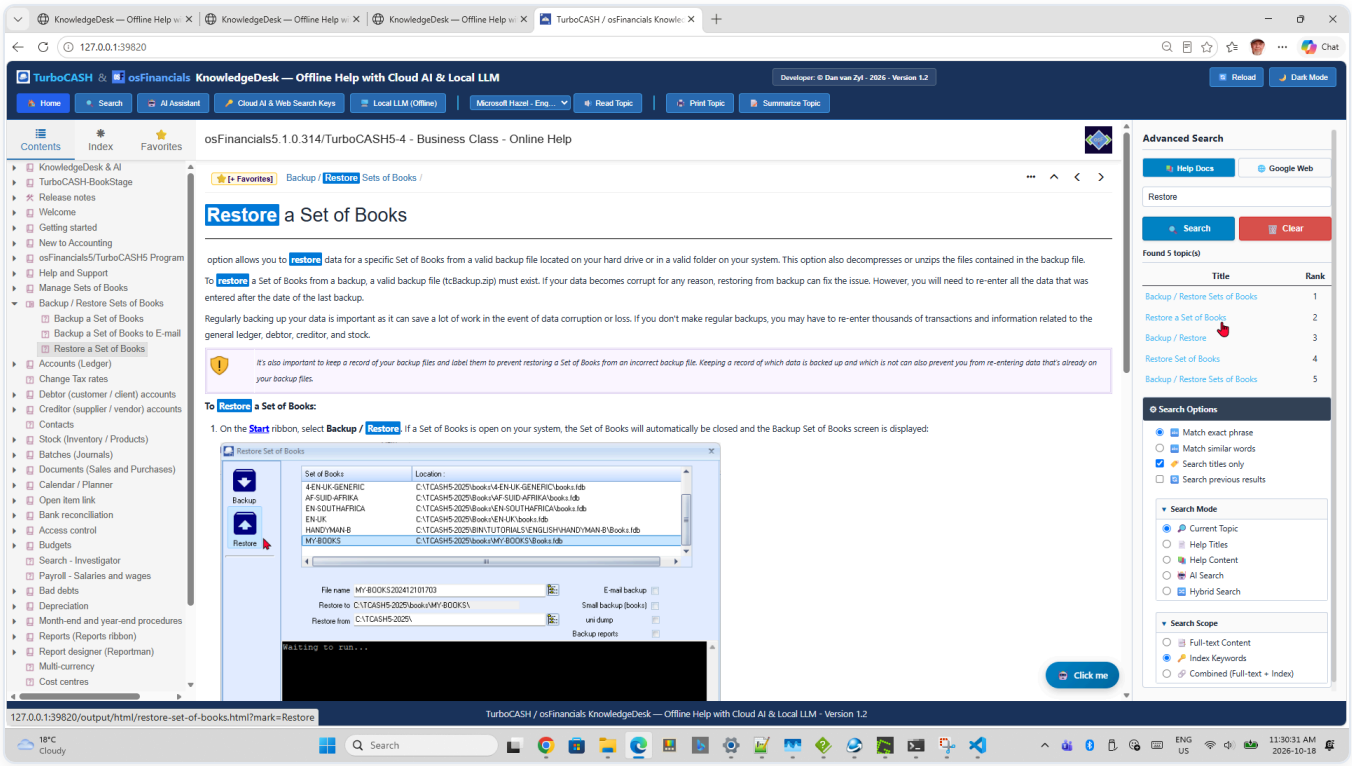
WHAT CHANGES BETWEEN THE TWO THEMES

ELEMENT	LIGHT MODE	DARK MODE
Page background	Light slate	Black
Panel background	White	Charcoal / Dark Grey
Body text	Near-black	Off-white
Headings (H1, H2, H3)	Dark navy	White
Active Left Pane Tab (Contents, Index, Favorites)	White background, dark blue text	Black background, bright blue text
Table Headers	Light grey background with dark text	Dark grey background with light text
Accent colour	Deep sky blue	Bright sky blue
Header button	🌙 Dark Mode	🌞 Light Mode

> 💡 **Note on TurboCASH / osFinancials Content:**
> The original TurboCASH / osFinancials software (including the standalone `EngHelp.chm` file) does **not** support Dark Mode.
>
> However, when viewing those same help topics inside KnowledgeDesk, the app's Dark Mode applies to **both the interface** (header, navigation bars, search panels) **and the documentation content topics** themselves. This provides a completely unified dark reading experience.

> 💡 **The setting is per-browser, per-install.**
> Dark mode is saved to your browser's local storage for `http://127.0.0.1:39820`.
>
> It will **not** carry over if you:
>
> - Switch to a different browser (e.g., from Edge to Chrome)
> - Use a different install folder
> - Clear your browser's site data

4. Searching



Search results with autocomplete suggestions

4.1 — Quick Search

1. Click 🔍 **Search** in the navigation bar.
2. Type a keyword or phrase into the search box.
3. Press `Enter` or click the 🔍 **Search** button.
4. Click any result in the list to load that topic in the center pane.

4.2 — Autocomplete & typo correction

As you type, the search box suggests matching topics from the help index. Fuzzy matching handles minor typos — a search for *"jurnel"* will suggest *"Journal"*.

4.3 — Search Options (Filters)

Expand ⚙ **Search Options** below the search box to refine your query:

- 🔤 **Match exact phrase:** Require a letter-for-letter match.
- 🔤 **Match similar words:** Fuzzy matching (default) to handle typos and word variations.
- 🏷️ **Search titles only:** Restrict the search to topic headings, ignoring the body text.
- 🔄 **Search previous results:** Narrow down your search within the results of your current query.

4.4 — Search Mode

The **Search Mode** section lets you change *where* the search engine looks. This is useful if you want to search the current page, or if you want to use the built-in AI to find an answer.

- 🔎 **Current Topic:** Only searches within the help topic you currently have open in the center pane.
- 📄 **Help Titles:** Searches only the titles of the help topics.
- 📚 **Help Content:** Searches the full text of all help topics (the default mode).
- 🤖 **AI Search:** Uses your configured Cloud AI or Local LLM to find an answer based on the help content.
- 🔀 **Hybrid Search:** Combines traditional keyword searching with AI assistance for the most comprehensive results.

4.5 — Search Scope

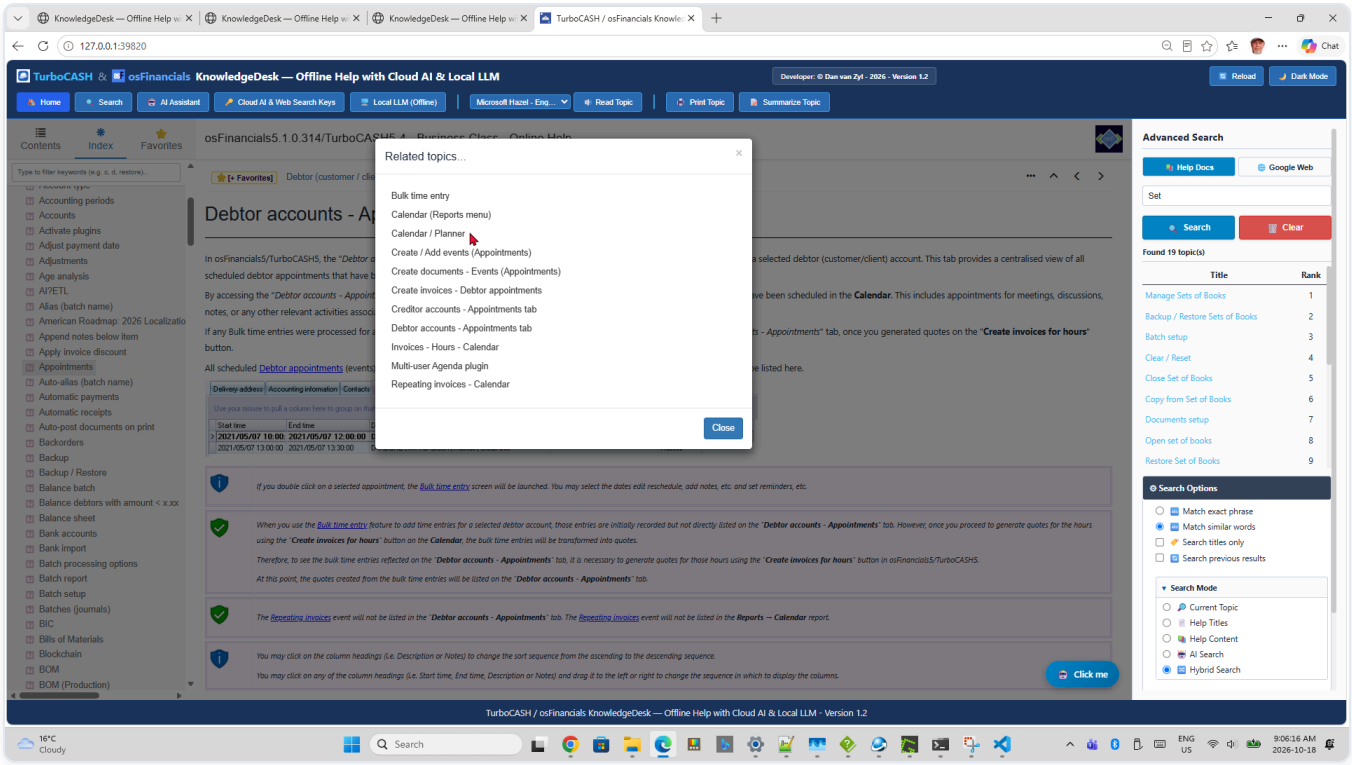
The **Search Scope** section defines how the search engine matches your keywords against the documentation database.

- 📑 **Full-text Content:** Searches every word inside the help topics. Best for finding specific phrases or paragraphs.
- 🔑 **Index Keywords:** Only searches the predefined keywords in the Index. Best for finding exact terminology.
- 🔗 **Combined (Full-text + Index):** Searches both the full text and the index keywords simultaneously (the default mode).

> 💡 **Index vs. Search — which to use?** Use the 🗂️ **Index** tab when you already know the exact term (it's instant, no ranking). Use 🔍 **Search** when you're exploring or only remember part of a phrase — it ranks results by relevance and tolerates typos.

4.4 — The Index tab

The 🗂️ **Index** tab in the left pane is a flat, alphabetical keyword list — the same index you would find at the back of a printed manual. It is the fastest way to jump straight to a topic when you already know the exact term (e.g. "*Debtors*", "*VAT*", "*Ledger*").



The Index tab — alphabetical keyword list with quick-jump letters

HOW TO USE THE INDEX

1. Click the **Index** tab in the left pane.
2. Type the first few letters of the keyword — the list scrolls directly to that section.
3. Click any entry to load the corresponding help topic in the center pane.

> 💡 **Related Topics Popups:** If a keyword in the Index has multiple associated topics, clicking it will open a **Related Topics...** popup screen. This popup lists all the matching topics, allowing you to select exactly the one you need.

> 💡 **Index vs. Search — which to use?** Use the **Index** when you already know the exact term (it's instant, no ranking). Use **Search** when you're exploring or only remember part of a phrase — it ranks results by relevance and tolerates typos.

5. Favorites & Bookmarks

Favorites allow you to save high-value reference pages so you can jump straight back to them without having to search again. But first, you need to find the topic you want to save.

5.1 — Locating a Topic to Bookmark

Before you can bookmark a topic, you need to open it in the center pane. There are three ways to find the topic you need:

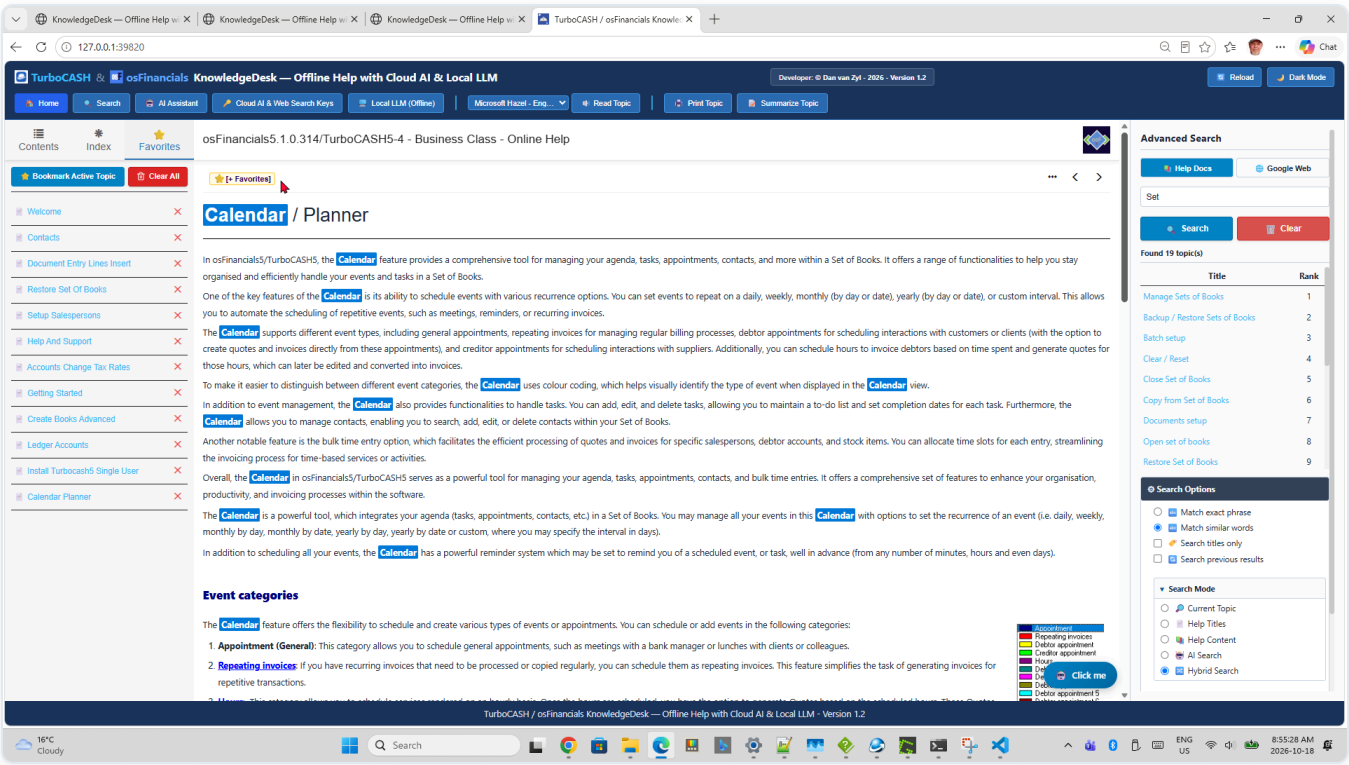
- 📚 **The Contents Tab:** Click through the hierarchical tree of chapters and topics in the left pane.
- 🗂️ **The Index Tab:** Type or scroll to a specific keyword in the alphabetical list.
- 🔍 **Advanced Search:** Use the search panel in the right pane to find specific phrases, which will display the matching topic in the center pane.

5.2 — Bookmarking a Topic

Once your desired topic is visible in the center pane, there are two ways to add it to your favorites:

- Click the ⭐ **[+ Favorites]** button located just above the topic title in the center pane.
- Open the ⭐ **Favorites** tab in the left pane and click the blue ⭐ **Bookmark Current Topic** button.

5.3 — Accessing Favorites



The Favorites tab — every bookmarked topic in one list

Click the ⭐ **Favorites** tab in the left pane to see all your saved bookmarks. Click any entry in the list to instantly load that topic in the center pane.

5.4 — Removing Favorites

You can remove bookmarks in two ways:

- Click the ✕ button next to any individual bookmark to remove just that topic.
- Click the **Clear All** button at the top of the Favorites tab to wipe the entire list instantly.

> 💾 **Where are my favorites stored?** Favorites are saved privately in your web browser's Local Storage for the address `http://127.0.0.1:39820`.
>
> They persist across restarts and survive app upgrades. However, they do **not** carry over if you:
>
> - Switch to a **different web browser** (e.g., from Edge to Chrome)
> - Clear your browser's site data or cache
> - Change the port the app runs on

6. AI Features

KnowledgeDesk — Offline Help with Cloud AI & Local LLM can generate summaries and answer questions using either cloud AI or a local model running on your machine.

6.1 — Three-Tier Routing

When you click 📝 **Summarize Page** or ask the 🤖 **AI Assistant**, the app automatically picks the best available AI mode:

1. 🖥️ **Local LLM** — if the toggle is ON and a model is selected
2. ☁️ **Cloud AI** — if a cloud API key is saved
3. 📄 **Offline extract** — always available as fallback

Each tier is designed for a specific use case. The active mode is announced in the response header so you always know which engine produced the answer.

MODE	HEADER FORMAT	BEST FOR
🖥️ Local LLM	`🖥️ Local LLM Summary: [Topic]` (model name + time shown below)	Privacy-focused work; offline environments
☁️ Cloud AI	`📝 AI Summary (PROVIDER): [Topic]`	Fast, high-quality answers; broadest knowledge
📄 Offline extract	`📄 Offline Summary: [Topic]`	Instant answers with no setup — always available

> 💡 **The provider name is always shown.** For Cloud AI summaries the header names the specific provider that answered — `GEMINI`, `GROQ`, `OPENAI`, `DEEPSEEK`, `MISTRAL`, `COHERE`, or `OPENROUTER`. For Local LLM summaries the model name is shown (for example `llama3.2:3b`). This makes it easy to verify which service handled your request — useful when you have multiple keys saved.

ACCOUNTING-FOCUSED ANSWERS

Every AI mode is tuned for the TurboCASH / osFinancials accounting environment. Prompts sent to Cloud AI and Local LLM include a system instruction that establishes the app as an accounting assistant and asks for concise, practical answers with exact menu paths and accounting terminology. This makes the responses noticeably more relevant than a generic AI chat.

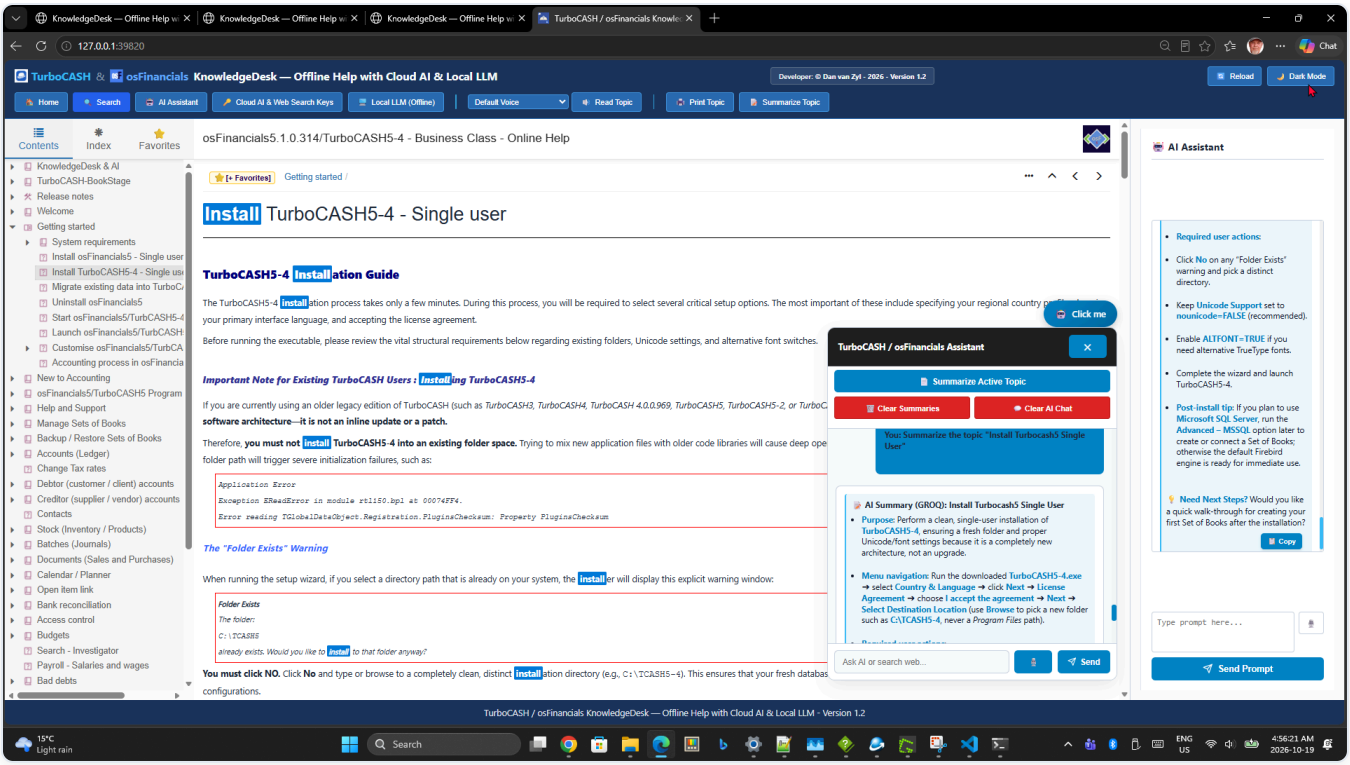
6.2 — Response badges

Every AI response shows a colored badge indicating its source:

BADGE	SOURCE
🖥️ `Local LLM Summary: [Topic]`	Generated by your local model. The model name and elapsed time appear just below the header.
📝 `AI Summary (PROVIDER): [Topic]`	Generated by a cloud provider. The specific provider that answered is shown in parentheses — for example `(GROQ)`, `(GEMINI)`, or `(OPENAI)`.
📄 `Offline Summary: [Topic]`	Keyword extract from the local help index — no AI involved, works instantly.

Summaries use the 📝 emoji for the AI Mode badge. Chat replies use ☁️ `Assistant (AI Mode · PROVIDER):` — the provider name is shown after a middle dot.

6.3 — The AI Assistant chat



Chat interface with an AI-generated topic summary

Click **AI Assistant** in the navigation bar to open the chat panel. Type a question and press `Enter`.

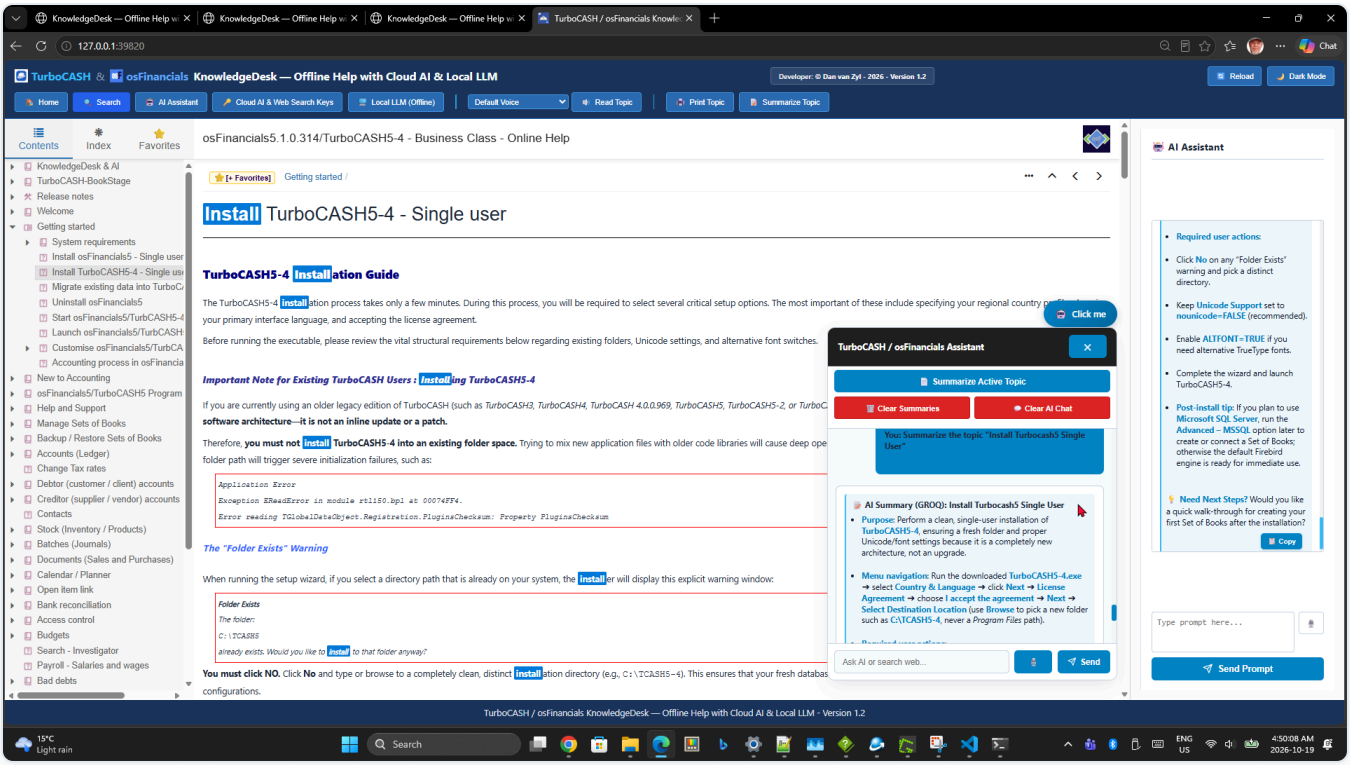
The assistant understands TurboCASH and osFinancials terminology. It will:

- Answer questions about menu paths and workflows
- Explain accounting concepts
- Suggest next steps for common tasks

> ⚠️ **First-time delay:** the first local LLM request after Ollama starts takes 60–140 seconds (the model loads into RAM). Subsequent requests are 5–20 seconds.

6.4 — Summarize Topic in detail

The **Summarize Topic** panel has three visually distinct regions. Understanding them makes the different AI modes much easier to follow.



The Summarize Topic panel showing a local summary, AI output, and the reset controls.

WHAT YOU SEE ON THIS SCREEN:

1. **Summarize Active Topic (Button)**
   Triggers a summary of the topic currently loaded in the centre pane. It works offline from local documentation, but will automatically upgrade to AI Mode (Cloud or Local LLM) if configured.
2. **Topic Summary Card**
   The generated bullet-point summary. Local mode builds it from the documentation index. AI Mode rebuilds it with your configured Cloud AI provider. Local LLM mode uses Ollama on your own machine.
3. **Clear Summaries / Clear AI Chat (Buttons)**
   Two independent reset buttons. *Clear Summaries* wipes every cached summary. *Clear AI Chat* wipes only the AI Assistant conversation history.
4. **AI Assistant Pane + 📋 Copy**
   Shows the AI Mode or Local LLM output. Press the 📋 **Copy** button in this header to send the summary straight to your clipboard, so you can paste it into Word, Outlook, Teams, or a helpdesk ticket.

LOCAL SUMMARY VS. AI MODE SUMMARY

Without an API key, the summary is built from the local help documentation — fast, offline, and no cost. As soon as any Cloud AI key is saved (see Section 7), the same button produces an AI-generated summary instead. The two are visually similar, so watch for the *Assistant (AI Mode)* badge in the right-hand pane to confirm which mode produced the output.

> ⚠️ **Clear Summaries vs. Clear AI Chat are not the same.** If your AI chat feels "stuck" or repeats stale context, use **Clear AI Chat**. If a topic summary looks wrong or outdated after the documentation is updated, use **Clear Summaries** — it forces every topic to be summarised again from scratch.

6.5 — Copying summaries and AI results to your favorite app

Every summary produced by KnowledgeDesk can be copied straight into your favorite application — Word, Outlook, OneNote, Teams, a chat window, a ticketing system, or another AI tool. There is no lock-in: the text is plain, selectable, and clipboard-ready.

THREE SOURCES, ONE CLIPBOARD

SOURCE	WHAT GETS COPIED	WORKS OFFLINE?
📄 **Local documentation summary**	The bullet-point Topic Summary built from the help index	✅ Yes — no key, no internet
🤖 **AI Mode summary**	The AI-generated summary from your configured Cloud AI provider (Gemini, OpenAI, Claude, etc.)	❌ Requires an API key
🖥️ **Local LLM summary**	The AI-generated summary from Ollama running on your own machine	✅ Yes — fully private, no cloud

HOW TO COPY

1. Run 📝 **Summarize Topic** on any topic, or ask the 🤖 **AI Assistant** a question.
2. In the panel header, click 📋 **Copy** (marker ④ above).
3. Switch to your favorite app and paste with `Ctrl` + `V`.

You can also select any portion of the summary with the mouse and use `Ctrl` + `C` — the panel is a normal selectable text region, so partial copies work too.

> 💡 **Perfect for email and tickets.** Summaries paste cleanly into Outlook, Gmail, Teams, Slack, Jira, ServiceNow, or a Word document — no images, no formatting baggage, just readable text. This makes KnowledgeDesk a fast "research → paste" tool for support staff.

📋 **What gets copied?**
When you click the 📋 **Copy** button, the clipboard captures the entire response, including the **Source Header**.

For example, if you copy a Local LLM summary, the text will look like this:

```
🖥️ Local LLM Summary: Contacts
qwen2.5:3b • 23.8s

The Contacts feature in osFinancials5/TurboCASH5...
```

This ensures that when you paste the text into an email or a support ticket, the recipient knows exactly where the information came from and which engine generated it.

✅ **Private by default.** Local documentation summaries and Local LLM summaries never leave your machine. Only AI Mode summaries (Cloud AI keys) send the topic text to the provider you configured — and only when you press Summarize or Send.

6.6 — What works with and without an API key

KnowledgeDesk blurs the line between "help system" and "AI assistant", and that's a good thing — but it also means the exact same button can behave differently depending on whether a key is configured. This section clarifies what works out of the box and what needs a key.

SUMMARY OF BEHAVIOURS

ACTION	API KEY NEEDED?	WHAT YOU GET WITHOUT A KEY
📝 **Summarize Topic** (local docs)	❌ No	Bullet-point summary extracted from the local help index — instant, offline.
🤖 **AI Assistant** — first question	✅ Yes	*"No active AI API key found."* The tip suggests entering a Groq, Gemini, DeepSeek, or OpenAI key in the Cloud AI settings.
💬 **Follow-up prompts** ("Yes please?", "Explain more", "Give an example")	✅ Yes	
📋 **Copy** a local summary	❌ No	Works — the local summary is copyable regardless of AI configuration.
📋 **Copy** an AI Mode summary	✅ Yes (to generate)	Nothing to copy until a key is saved and a summary is generated.

WHY FOLLOW-UP PROMPTS REQUIRE A KEY (OR LOCAL LLM)

The local documentation engine can extract keywords and bullet points from a topic — that's a deterministic, offline operation. But **follow-up prompts require conversational memory**: the assistant has to remember what you just asked, reason about it, and produce a fresh answer.

In v1.2, **Cloud AI** and **Local LLM** have this memory built-in. However, the **Offline Extract** mode does not. It is a one-shot keyword extractor, not a chatbot.

> 💡 **Rule of thumb**
> **One-shot summaries** of individual topics work offline.
> **Any back-and-forth conversation** needs a key (Cloud AI) *or* a running Local LLM.

WHAT THE "NO ACTIVE AI API KEY FOUND" MESSAGE MEANS

If you type a follow-up prompt and see this message in the AI Assistant pane:

> 🤖 **AI Assistant**
>
> ---
>
> ⚠️ **No active AI API key found.**
>
> 💡 *Need follow-up questions? Add a key in* 🔑 ***Cloud AI & Web Search Keys*** *or enable* 🖥️ ***Local LLM (Offline)*** *for conversational summaries and step-by-step guidance.*
>
> **Quick Setup:**
> • Add a free **Groq** API key (Recommended)
> • Or turn ON **Local LLM** if you have Ollama installed.

...it means one of three things:

1. **No Cloud AI key has been saved yet** — the most common cause. See Section 7 to add one (we recommend **Groq** for speed).
2. **A key was saved but the app hasn't been reloaded** — the app reads keys at startup. Reload the page after saving.
3. **Local LLM is OFF and no cloud key exists** — either turn on the Local LLM toggle (see Section 8) or save a cloud key.

ENABLING FOLLOW-UP QUESTIONS

Two paths — pick whichever fits your setup:

PATH	SETUP EFFORT	BEST FOR
☁️ **Add a Cloud AI key** Groq, Gemini, Mistral, OpenAI, Anthropic, DeepSeek, OpenRouter	Sign up for a free key, paste it, reload.	Most users — 5 minutes setup, works immediately.
🖥️ **Enable Local LLM** Ollama + a pulled model	Install Ollama, pull a model, set `OLLAMA_ORIGINS`, toggle ON.	Privacy-focused users, offline environments.

> ⚠️ **A saved key is not enough on its own.** After pasting a key and clicking 💾 **Save Keys**, you must **reload the app** — keys are read at page load. If you skip the reload, follow-up prompts keep returning the "No active AI API key found" message even though the key is stored.

> ✅ **Free options exist. Groq** and **Gemini** both offer a free tier with no credit card required — good enough for typical KnowledgeDesk use. (Note: **Groq is the default** in KnowledgeDesk due to its faster speeds and fewer server errors). See Section 7 for sign-up links to every provider, including free, freemium, and paid tiers.

6.7 — Why some replies ask follow-up questions

The app uses a simple rule to decide whether to offer a follow-up question at the end of a reply:

MODE	FOLLOW-UP OFFERED?	WHY
📄 Offline extract	❌ No	There is no AI to answer a follow-up. Instead, the footer suggests two ways to enable AI.
☁️ Cloud AI	✅ Yes	The provider can answer — replies end with a "💡 Need Next Steps?" question.
🖥️ Local LLM	✅ Yes	The local model can answer — replies end with a "💡 Need Next Steps?" question.

> 💡 **Why does this matter?** On a fresh install with no AI configured, the offline summary ends with a friendly nudge — "*Need follow-up questions? Add a key in 🔑 Cloud AI & Web Search Keys or enable 🖥️ Local LLM (Offline)...*" — rather than a question you can't answer. Once you save a Cloud AI key or enable Local LLM, the summary starts offering "💡 Need Next Steps?" questions that lead somewhere useful.

WHAT HAPPENS WHEN YOU CLICK A FOLLOW-UP

When you reply to a "Need Next Steps?" question — for example typing "*Yes please?*" or "*Explain more*" — the app has full context of the topic and the previous reply. Cloud AI and Local LLM both remember the last few exchanges, so follow-ups are coherent and on-topic.

6.8 — Table rendering in AI replies

Cloud AI and Local LLM models frequently return structured data as markdown tables — pipe-delimited rows that look like this in raw form:

```
| Field        | What to enter                              |
|--------------|--------------------------------------------|
| Contact Name | Full name of the person or entity          |
| Account      | Choose the debtor or creditor account      |
| Default      | Tick to auto-select on sales/purchase docs |
```

In previous versions these pipes were displayed as raw text and were hard to read. In v1.2, **every AI-generated markdown table renders as a real HTML table** with a shaded header row, visible borders, and properly aligned columns.

WHAT THE SAME TABLE LOOKS LIKE NOW

FIELD	WHAT TO ENTER
Contact Name	Full name of the person or entity
Account	Choose the debtor or creditor account
Default	Tick to auto-select on sales/purchase docs

TABLE STYLING ADAPTS TO YOUR THEME

Tables render correctly in both **Light Mode** and **Dark Mode**. Column headers, borders, and cell backgrounds use theme-aware CSS variables, so the table is always legible — no washed-out text in dark mode, no ink-black rows in light mode.

ELEMENT	LIGHT MODE	DARK MODE
Table background	White	Charcoal
Header row	Light slate	Dark slate
Borders	Light grey	Medium grey
Body text	Near-black	Off-white

WHAT GETS FORMATTED INSIDE A CELL

Even inside table cells, the AI's markdown emphasis is preserved:

- `**bold**` renders as **bold**
- `*italic*` renders as *italic*
- `` `code` `` renders as `code`
- Long cell content wraps naturally instead of overflowing
- Rows with fewer cells than the header are padded with empty cells so the table stays aligned

> 💡 **Which replies contain tables?** Most often:
>
> - Field-by-field descriptions ("what to enter in each field")
> - Menu-path cheat sheets
> - Comparison tables (e.g. "which tax rate applies when")
> - Troubleshooting tables (symptom / cause / fix)
> - Quick reference sheets at the end of a long walk-through
>
> You'll see these especially from **Groq** and **Cohere**, which favor structured output.

COPYING A TABLE

The 📋 **Copy** button on any AI reply copies the whole response, including tables. Paste into Word, Outlook, Teams, or OneNote and the table structure is preserved as a real table — not as a wall of pipe characters. Paste into a plain-text editor and you get a clean plain-text version instead.

CHECKBOX LISTS IN AI REPLIES

Related to tables, the AI sometimes returns task-list syntax:

```
- [ ] Open Contacts via Home ➔ Contacts.
- [ ] Add any new contacts needed.
- [x] Save the Chart of Accounts.
```

In v1.2 these render as proper checkbox glyphs:

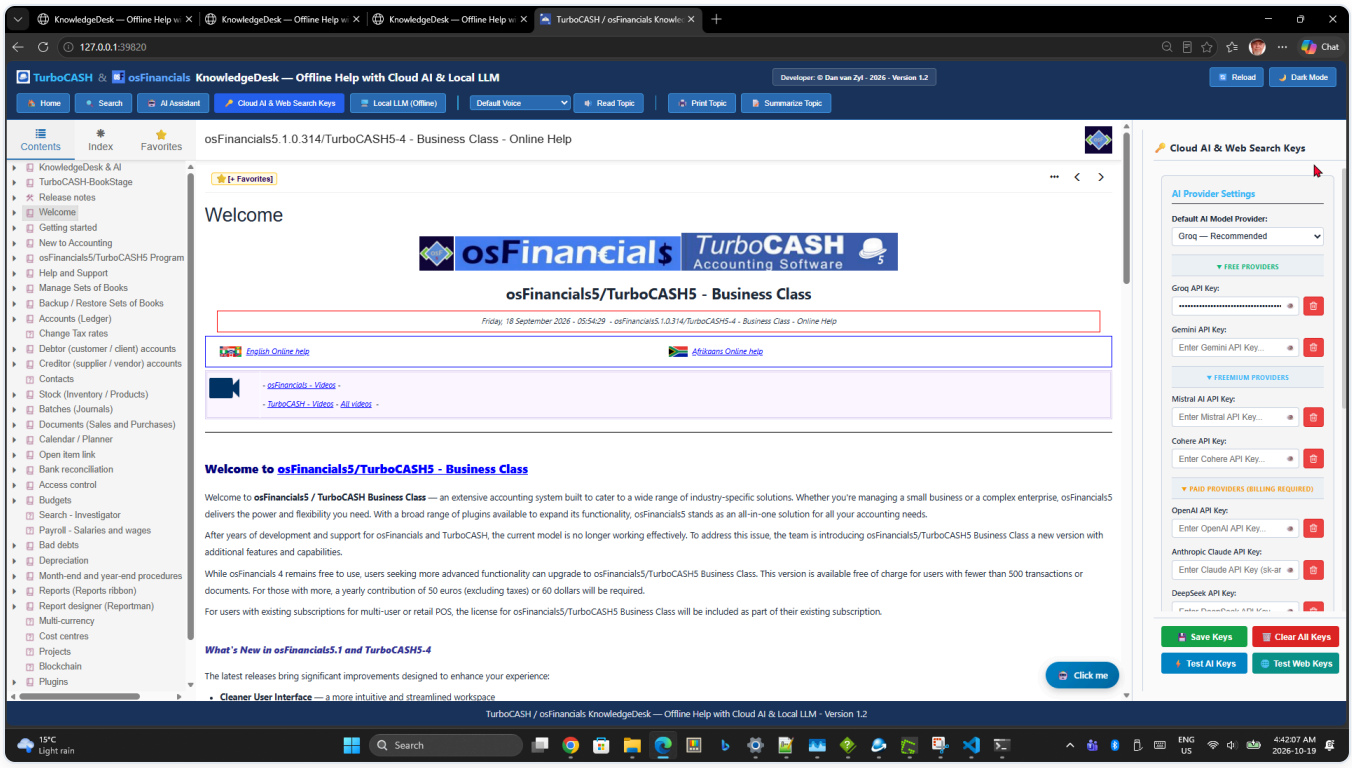
☐ Open Contacts via Home ➔ Contacts.

☐ Add any new contacts needed.

☑ Save the Chart of Accounts.

This makes AI-generated checklists immediately usable as to-do lists, without any cleanup.

7. Cloud AI Configuration



Cloud AI & Web Search Keys panel

7.1 — Where to get an API key

Free providers are best to start with:

PROVIDER	COST	SIGN-UP LINK	NOTES
Groq	Free tier	console.groq.com/keys	Fast inference; reliable for testing. Models rotate periodically. **Recommended default.**
Google Gemini	Free tier	aistudio.google.com/app/apikey	Generous free tier; occasionally rate-limited during peak demand. Accepts both `AIza...` and newer `AQ...` key formats.
Mistral AI	Freemium	console.mistral.ai/api-keys	Free tier grants access to smaller models. The app tries several models in order to find one your tier allows.
Cohere	Freemium	dashboard.cohere.com/api-keys	Trial key: ~1,000 API calls/month. Uses dated model names (e.g., `command-a-03-2025`).
OpenRouter	Freemium	openrouter.ai/keys	Meta-provider with free-tier models. Rate limit: ~50 requests/day without a deposit.
OpenAI	Paid	platform.openai.com/api-keys	Requires a payment method and credit balance.
Anthropic Claude	Paid	console.anthropic.com/settings/keys	Requires a payment method and credit balance.
DeepSeek	Paid	platform.deepseek.com/api_keys	Requires a top-up; low per-token cost. Also used for some Chinese-language models.

7.2 — Saving a Key

1. Click 🔑 **Cloud AI & Web Search Keys** in the navigation bar.
2. Locate the field for your chosen provider (e.g., *Groq API Key*, *Gemini API Key*).
3. Paste your API key into the field. The characters are masked as `••••••••` while you type.
4. **To verify the key was entered correctly:** click the 👁 **eye icon** to the right of the field. The masked dots become the plain-text key so you can visually confirm the paste (no truncation, no trailing spaces). Click the eye again (now shown as 🙈) to re-mask the field.
   Tip: use this only when you suspect a paste went wrong. In normal use, keep the field masked.
5. Optionally, if you have multiple keys saved, choose the provider you want to use by default from the **Default AI Model Provider** dropdown at the top of the panel.
6. Click 💾 **Save Keys**.
7. A confirmation dialog appears. The dialog color tells you whether validation passed:
   - **Green (Success)** — every saved key passed format validation and will work when used.
   - **Amber (Warning)** — one or more keys did not match the expected format for their provider. The dialog lists the valid and invalid keys separately so you can fix the problem immediately.
8. Example of a green (all-valid) dialog:

> ✅ **Saved 2 keys:**
> - Groq (Llama 3 Fast Inference)

> - Google Gemini
>
> **Default model:** GROQ

9. Example of an amber (validation warning) dialog:

> ⚠️ **Saved 3 keys — with validation warnings:**
>
> ✅ **Valid format:**
> - Groq (Llama 3 Fast Inference)
>
> ⚠️ **Invalid format:**
> - Google Gemini
> - OpenAI
>
> *The invalid keys are saved, but they will not work when used. Check that each key is in the correct field for its provider.*
>
> **Default model:** GROQ

10. Click **OK** to dismiss the dialog.
11. **Optional but recommended:** reload the app (`Ctrl+Shift+R`) to confirm the key was stored correctly. The browser console will log `[Vault] Active providers: ...` at startup, listing every provider whose key is currently saved.

> **About the Default AI Model Provider dropdown:** when two or more providers have saved keys, this dropdown decides which one handles summaries and chat replies. Each reply's header shows the provider that was used — for example `Assistant (AI Mode · GROQ):` . `AI Summary (GEMINI):` or If the selected provider is unavailable, the router automatically falls back to another provider that has a valid key.

The default model selector at the top of the panel controls which provider is used when multiple keys are saved.

7.3 — Testing, clearing, and deleting keys

Below the API key fields, three buttons let you verify and manage your saved keys. Each opens a styled modal dialog with an OK button.

BUTTON	WHAT IT DOES
💾 **Save Keys**	Saves every API key entered and the selected default model. Validates each key's format and reports the results — see Section 7.2.
⚡ **Test AI Keys**	Reports the validation status of **all 8 AI providers**, not just the ones with saved keys. Useful for a full status snapshot — you'll see Valid ✅, Invalid format ⚠️, or Missing ❌ for each provider, plus the current default model.
🌐 **Test Web Keys**	Reports the status of the Web Search credentials — Bing API Key and Google CSE ID. Confirms whether web-grounded AI results are enabled.
🗑️ **Clear All Keys**	Permanently removes every saved key. A confirmation dialog *lists every provider that will be cleared* before you commit. Resets the default model to **Groq**.

FORMAT VALIDATION ON SAVE AND TEST

Both 💾 **Save Keys** and ⚡ **Test AI Keys** run the same format validator. Each provider has its own expected key prefix:

PROVIDER	EXPECTED KEY FORMAT
Google Gemini	Starts with `AIza` or `AQ.`
Groq	Starts with `gsk_`
Anthropic Claude	Starts with `sk-ant-`
OpenRouter	Starts with `sk-or-`
OpenAI / DeepSeek	Starts with `sk-` (ambiguous — both use this)
Mistral / Cohere	Length + no foreign prefix (the app rejects keys clearly meant for other providers)
Bing API Key	Length + no foreign prefix

The validator rejects keys pasted into the wrong field. For example, a Groq key (`gsk_…`) pasted into the Gemini field will be reported as **Invalid format** ⚠️ — not as a valid Gemini key. This catches cross-provider paste mistakes before they cause a failed request later.

DELETING AN INDIVIDUAL KEY

Each API key field has its own 🗑️ button to the right. Clicking it opens a confirmation dialog:

1. Click the 🗑️ button next to the provider whose key you want to remove.
2. A dialog appears: "*Delete the stored [Provider] key?*" with **Cancel** and **Delete** buttons.

3. Click **Cancel** to keep the key, or **Delete** to remove it.
4. On delete, a green success message confirms "🗑️ *[Provider] key removed.*"

> 💡 **Cancel truly cancels.** The key is not removed until you click Delete in the confirmation dialog. If you change your mind, Cancel is safe.

7.4 — Model Fallback Chains

Every cloud provider publishes a catalog of models. Providers occasionally retire models, restrict them by subscription tier, or experience temporary capacity issues on a specific model. To keep AI features working even when this happens, KnowledgeDesk tries **multiple models per provider** in sequence.

HOW IT WORKS

1. You click 📝 **Summarize Topic** or send a chat message.
2. The app picks the provider you selected in the **Default AI Model Provider** dropdown.
3. It tries the provider's **first model** in its list.
4. If the model is unavailable — a 403 tier restriction, a 404 deprecated model, a 429 rate limit, or a 5xx service error — the app tries the **next model** in the chain.
5. This continues until a model succeeds or the chain is exhausted.
6. If all models fail, the reply shows the last error and names the provider in the header.

MODEL CHAINS PER PROVIDER

PROVIDER	MODEL FALLBACK ORDER	TYPICAL TRIGGER FOR FALLBACK
Google Gemini	`gemini-3.8-flash` → `gemini-3.7-flash` → `gemini-3.6-flash`	503 "high demand" on newer models
Groq	`openai/gpt-oss-120b` → `openai/gpt-oss-20b`	Model availability, capacity
Mistral AI	`mistral-large-latest` → `mistral-medium-latest` → `mistral-small-latest` → `open-mistral-7b` → `open-mixtral-8x7b`	403 "not in subscription tier" on paid models
Cohere	`command-a-03-2025` → `command-r7b-12-2024` → `command-r-plus-08-2024` → `command-r-08-2024`	Model deprecation (Cohere rotates its catalog annually)
DeepSeek	`deepseek-chat` → `deepseek-reasoner` *(where available)*	Region or tier availability
OpenAI	`gpt-4o-mini` → `gpt-4o` *(if available)*	Model or account restrictions
OpenRouter	`meta-llama/llama-3.3-70b-instruct:free` → `mistralai/mistral-7b-instruct:free` → `openai/gpt-4o-mini`	Free tier exhaustion, upstream outage

> 💡 **What a fallback looks like in the console.** When a fallback happens, the browser console logs which model failed and which one will be tried next. For example:
>
> ```
> Gemini model gemini-3.8-flash failed: This model is currently experiencing high demand...
>   → Trying next...
> Gemini model gemini-3.7-flash failed: This model is currently experiencing high demand...
>   → Trying next...
> ```
>
> You'll see the successful response appear in the chat panel once a working model answers.

WHY FALLBACK MATTERS

- **No dead ends.** A single overloaded model doesn't take the whole provider offline for you.
- **Tier flexibility.** Free-tier users can still use providers like Mistral, even if the paid model is unavailable — the app silently routes to a smaller model their tier allows.
- **Deprecation resilience.** When a provider retires a model (Cohere's `command-r-plus`, for example), the app tries the newer replacements automatically.
- **Zero configuration.** Users don't need to know which model is being used — the header on the reply always names the provider that answered.

> ✅ **The header always tells the truth.** Even if the fallback chain is running, the reply header shows the provider that finally succeeded — `📝 AI Summary (MISTRAL):` or `Assistant (AI Mode · GEMINI):`. There is never a mismatch between the label and the actual source.

WHAT HAPPENS WHEN EVERY MODEL FAILS

If the fallback chain is exhausted (rare — happens only during a full provider outage), the app returns a message like:

```
📝 AI Summary (GROQ): [Topic]
Groq Error: [last error message from the final attempt]
```

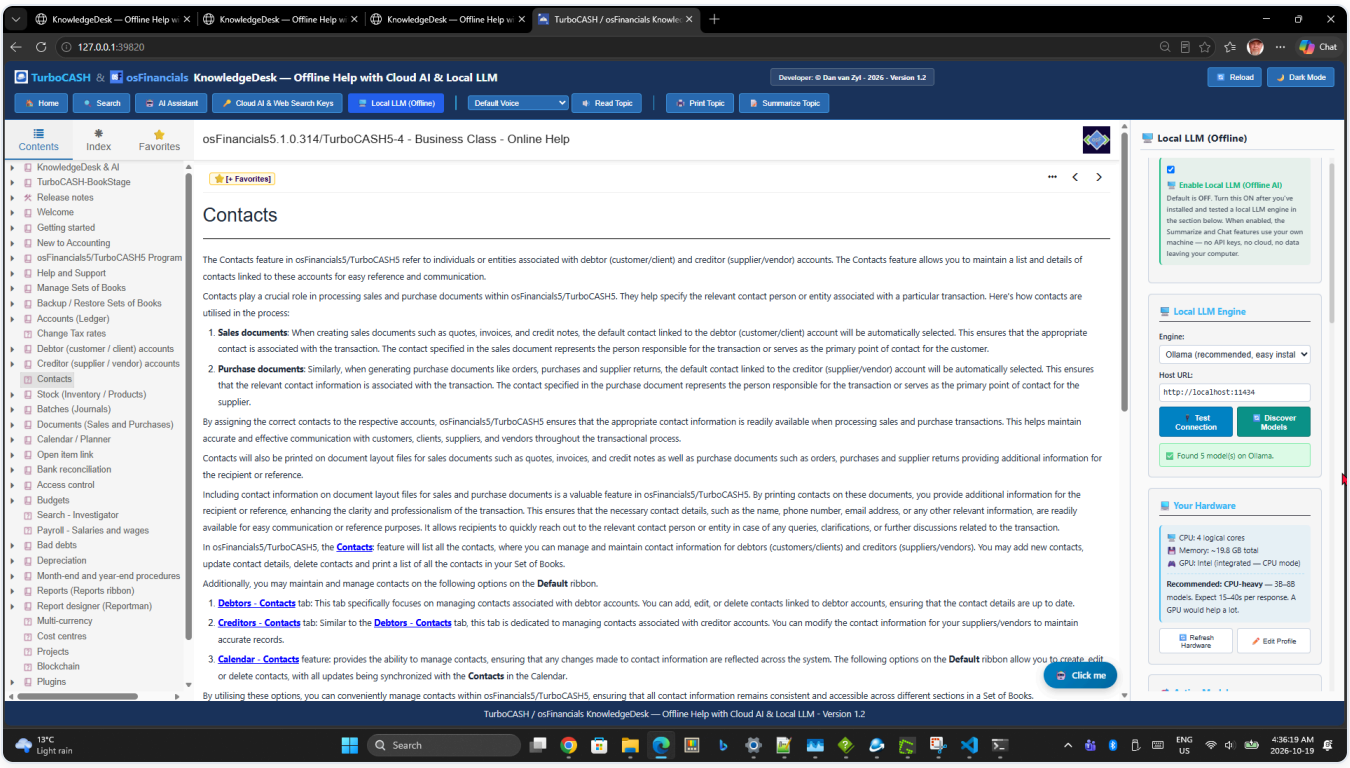
The header still names the provider that was attempted, and the error body tells you what went wrong. You can:

- **Wait a few minutes** — most 503 errors clear on their own
- **Switch provider** using the Default AI Model Provider dropdown
- **Enable Local LLM** if you have Ollama installed — it works entirely offline

ADDING NEW MODELS TO THE FALLBACK CHAIN

If a provider retires a model that the app still calls (rare, but happens — most recently with Groq's `llama-3.3-70b-versatile` in August 2026), the fallback chain will keep trying the retired model as one of its attempts. To update the chain, edit `vault.js` → `VAULT.CLOUD_MODELS` → the provider's `models` array. That is the single source of truth for the whole app.

8. Local LLM (Offline AI)



Local LLM panel — engine, models, hardware detection, toggle

8.1 — Why use local LLM?

- **Privacy** — nothing leaves your computer
- **No API keys** — no sign-ups, no cloud accounts
- **Offline** — works without internet
- **Free** — no per-request charges

8.2 — Panel Overview & Setup

To use offline AI, you need an engine installed on your machine. **Ollama** is recommended because it is easy to install.

STEP 1: INSTALL OLLAMA & DISCOVER MODELS

1. Download from ollama.com/download
2. Install and run it (it starts automatically on port 11434).
3. Open a terminal and pull a model: `ollama pull qwen2.5:3b`
4. Return to KnowledgeDesk and click 🔄 **Discover Models**. This will populate the Local LLM models installed on your system (e.g., "*Found 5 model(s) on Ollama.*"). You may then select the appropriate LLM model from the 🎯 **Active Model** dropdown and 📦 **Available Models** list.

STEP 2: CONFIGURE THE PANEL

Open the 🖥️ **Local LLM (Offline)** panel in the navigation bar. You will see the following sections:

- 🖥️ **Local LLM Engine:** Select *Ollama* (or LM Studio / LocalAI) and verify the Host URL (default is `http://localhost:11434`).
  - Click 🔌 **Test Connection** to verify the connection. A green success message will appear: "✅ *Connected to Ollama at http://localhost:11434 — 5 model(s) installed.*"
  - Click 🔄 **Discover Models** to pull the list of installed models into the dropdown.
- 🖥️ **Your Hardware:** KnowledgeDesk automatically detects your machine's hardware to assist you in choosing the appropriate LLM model for your environment. It displays your CPU, total memory, and GPU.
  - If you have a GPU, it will say "*NVIDIA (8 GB VRAM)*". If you don't, it will say "*Intel (integrated — CPU mode)*".
  - It will give you a recommendation, such as: "*Recommended: CPU-heavy — 3B–8B models. Expect 15–40s per response. A GPU would help a lot.*"
  - Click 🔄 **Refresh Hardware** if you upgrade your PC components.
- 🎯 **Active Model:** Shows the model currently selected (e.g., `qwen2.5:3b (1.8 GB)`).
- 📦 **Available Models (by size):** Lists the models you have downloaded in Ollama, grouped by size (Tiny, Small, Medium, etc.).

> 💡 **Why is my RAM showing as 16 GB when I have 19.8 GB?**
> For privacy and security reasons, web browsers intentionally round system RAM to the nearest power of 2 (e.g., 8, 16, 32 GB).
>
> If your machine's hardware is not correctly detected, click the 🖍️ **Edit Profile** button. You can manually enter your exact total system RAM in GB (e.g., `19.8`). This ensures the app recommends the right size of AI model for your specific machine.

STEP 3: ENABLE THE TOGGLE

The **Enable Local LLM (Offline AI)** checkbox at the top of the panel is **OFF by default**. Turn it ON once your engine is running and a model is selected.

> ✅ **Why is it off by default?** New users who haven't installed Ollama yet won't see "failed to fetch" errors. The toggle is a controlled opt-in.

> ⚠️ **CORS Error?**
> If Discovery or Connection tests fail with a network error, Ollama is blocking browser access. You must set the environment variable `OLLAMA_ORIGINS=http://127.0.0.1:39820` and restart Ollama. (See Section 10 — Troubleshooting for step-by-step instructions).

8.3 — Enabling the toggle

The **Enable Local LLM (Offline AI)** checkbox at the top of the Local LLM panel is **OFF by default**. Turn it ON once your engine is running and a model is selected.

> ✅ **Why is it off by default?** New users who haven't installed Ollama yet won't see "failed to fetch" errors. The toggle is a controlled opt-in.

HOW THE LOCAL LLM INTEGRATES WITH SUMMARIES & CHAT

- **If Local LLM is ON:** It will use your installed models in the **Summarize Topic** and **AI Assistant** features to generate local, private AI replies.
- **If Local LLM is OFF:** It will fall back to the 📄 **Offline Summary** option (which is instant and built into the app).
- **If Local LLM times out:** If you have saved API keys for a Cloud AI provider (like Groq), the app will automatically use that cloud provider as a fallback so you still get an answer.

8.4 — Model recommendations

The single biggest factor in local LLM performance is the **model size**. Larger models produce better answers but take dramatically longer to run — especially on CPU-only machines. Pick a model that matches your hardware.

> ⚠️ **The rule of thumb.** On a machine **without a GPU**, stay at **3B parameters or smaller**. A 7B model will *time out* before it finishes a single summary. This is not a bug — it is the physics of running a large model on a CPU.

RECOMMENDED MODELS BY HARDWARE

HARDWARE	RECOMMENDED MODEL	EXPECTED WAIT
CPU-only 8 GB RAM or less	`qwen2.5:3b` or `llama3.2:3b`	90–150 seconds per summary
CPU-only 16 GB RAM	`qwen2.5:3b` *or* `llama3.2:1b` for speed	60–150 seconds per summary
NVIDIA GPU 6 GB VRAM or more	Up to `qwen2.5:7b` or `llama3.1:8b`	5–20 seconds per summary
Apple Silicon M1 / M2 / M3 with 16 GB+	Up to `qwen2.5:7b` or `llama3.1:8b`	10–30 seconds per summary

> ⚠️ **Choose a "Chat" or "Instruct" model, NOT a "Coder" model!**
> When downloading models from the Ollama library, you must choose a general-purpose model (like `qwen2.5:3b` or `llama3.2:3b`).
>
> If you download a **Coder** model (like `deepseek-coder`), the AI will try to write Python code instead of summarizing your accounting documentation, or it will say *"I don't have access to that information."* Coder models are only good for programming, not for reading help files.

WHY 7B MODELS DON'T WORK ON CPU

A 7B-parameter model like `qwen2.5:7b` contains roughly 4.4 GB of weights. On CPU-only hardware, generating text from these weights runs at approximately **2–5 tokens per second**. A typical 300-token summary therefore needs **60–150 seconds of pure generation time** — plus:

- **Model load** — 30–60 seconds if the model isn't already warm in memory
- **Prompt processing** — 30–60 seconds for a 1,000-token prompt (the topic context + instructions)

Total: **2–5 minutes** for one summary. KnowledgeDesk allows 150 seconds before falling back to cloud AI — so a 7B model on CPU never finishes in time.

A 3B model such as `qwen2.5:3b` requires roughly **one third of that compute** for the same task, which puts it comfortably under the timeout.

WHAT HAPPENS WHEN THE LOCAL MODEL TIMES OUT

When a local LLM request exceeds its timeout, the app does not fail — it **falls back to cloud AI** if a cloud key is saved. This is why the reply header might show a different provider than the model you selected. You will see this in the browser console (F12):

```
[Summarize] Local LLM failed, falling back: Timeout after 150s
  — model may still be loading
```

The reply you see then comes from the cloud provider (Groq, Gemini, etc.), and the header reflects that: `📝 AI Summary (GROQ):` instead of `🖥️ Local LLM Summary:`.

> 💡 **The header always tells the truth.** If you see `📝 AI Summary (GROQ):` when you expected a local model, the console will show you why. Most often it's the timeout; occasionally it's Ollama not running or a network error. The header never lies about which engine produced the answer.

FULL MODEL SIZE REFERENCE

MODEL	SIZE	CPU-ONLY SPEED	QUALITY
`qwen2.5:0.5b`	0.4 GB	~30 seconds	Basic — good for simple Q&A
`llama3.2:1b`	1.3 GB	~60–80 seconds	OK — usable for help docs
`qwen2.5:1.5b`	1.0 GB	~60–90 seconds	OK — concise answers
`qwen2.5:3b`	1.8 GB	**~140 seconds** ✅	**Recommended** for CPU
`llama3.2:3b`	1.9 GB	~150 seconds ✅	Alternative to qwen2.5:3b
`qwen2.5:7b`	4.4 GB	~300 seconds ❌	Too slow on CPU
`llama3.1:8b`	4.9 GB	~400 seconds ❌	Requires a GPU

HOW TO SWITCH TO A SMALLER MODEL

1. Open 🖥️ **Local LLM (Offline)** in the navigation bar.
2. Click the 🎯 **Active Model** dropdown.
3. Choose `qwen2.5:3b` (recommended) or another 1B–3B model.
4. Close the panel. The selection takes effect immediately on the next request.

If the model you want isn't in the list, download it via `ollama pull qwen2.5:3b`, then click 🔄 **Discover Models** in the Local LLM panel to refresh the dropdown.

SUMMARY OF THE RULE

IF YOUR MACHINE HAS…	USE MODELS UP TO…
CPU only, ≤ 8 GB RAM	**3B parameters**
CPU only, 16 GB RAM	**3B parameters** (or 1B for speed)
NVIDIA GPU, 6–8 GB VRAM	7B parameters
NVIDIA GPU, 12 GB+ VRAM	13B parameters
Apple Silicon, 16 GB+ unified	7B parameters

When in doubt, start smaller. A 3B model that finishes in 2 minutes is far more useful than a 7B model that times out every time.

9. Reader Tools

9.1 — 🔊 Read Topic (Text-to-Speech)

Click 🔊 **Read Topic** to have the current topic read aloud. While playback is running, three additional controls appear:

- ⏸ **Pause** — pause playback
- ▶ **Resume** — continue from where paused
- ⏹ **Stop** — end playback and clear highlighting

The currently-read paragraph is highlighted in yellow. The highlight follows the text in sync with the spoken words.

CHOOSING A VOICE

Before clicking 🔊 **Read Topic**, use the **Default Voice** dropdown in the navigation bar to pick a specific voice. Every voice installed on your operating system and exposed by your browser appears in the list, tagged with its language code (e.g. `en-US`, `en-GB`, `af-ZA`). Male, female, and (in some cases) child voices are all available if the OS provides them.

VOICE AVAILABILITY BY BROWSER

The number of available voices depends on two things: your **browser** and your **operating system's installed speech packs**. Microsoft Edge ships with the largest built-in voice catalog by far.

BROWSER	VOICES AVAILABLE	NOTES
Microsoft Edge	**~337 voices** (male & female, many country dialects)	Best support. Uses Microsoft Azure neural voices provided by Windows. Full coverage of English (US, UK, AU, CA, IN, ZA, IE, NZ), plus dozens of other languages.
Google Chrome	**~25–40 voices**	Uses Google's online TTS engine. Much smaller catalog than Edge, and the exact count varies by platform.
Brave	Same as Chrome	Chromium-based, uses the same OS speech engine.
Firefox	❌ **Not supported**	Firefox does not expose the `speechSynthesis` API in a way the app can use. Clicking Read Topic shows a "text-to-speech is not supported" message. Use Edge or Chrome instead.
Safari	Limited / inconsistent	WebKit implementation varies; support is not guaranteed.

> ✅ **Best experience:** Microsoft Edge on Windows 10 / 11 gives you the largest and best sounding voice catalog — **337 voice variations** covering male and female speakers in dozens of regional accents (US, UK, Australia, Canada, India, South Africa, Ireland, New Zealand, and more). If voice quality matters to you, use Edge.

HOW TO CHECK WHICH VOICES YOU HAVE

1. Open the app in your browser.
2. Look at the **Default Voice** dropdown in the navigation bar.
3. Click it. The list shows every voice currently available — the count varies by browser and OS.

If you only see a handful of voices, install additional language packs:

- **Windows 11:** *Settings → Time & Language → Speech → Voices*. Add or remove installed voices.
- **Windows 10:** *Settings → Time & Language → Speech → Manage voices*.
- **macOS:** *System Settings → Accessibility → Spoken Content → System Voice → Manage Voices*.

USING THE VOICE SELECTOR

1. Open any help topic.
2. Pick a voice from the **Default Voice** dropdown.
3. Click 🔊 **Read Topic**.
4. The chosen voice reads the topic aloud. If you don't pick a voice, the browser uses its default for the OS language.

> 💡 **Voice, or nothing?** If the 🔊 **Read Topic** button does nothing in your browser, check:
>
> 1. Are you in Edge or Chrome? Firefox and Safari are not supported.
> 2. Is your system volume on and not muted?
> 3. Does the Default Voice dropdown list any voices? If not, install a speech pack via your OS settings.

9.2 — 🖨️ Print Topic

Click 🖨️ **Print Topic** to send the *currently active topic* to a printer or save as PDF. The topic is formatted for clean output: navigation bars, headers, sidebars, and UI controls are automatically removed, leaving only the core text and images.

> 💡 **Why use this?** Clean printouts are incredibly useful for:
>
> - **Customer Support:** Printing specific instructions to hand to a client.
> - **Training:** Creating quick reference guides for new staff.
> - **Feedback:** Sending a specific page to the development team to suggest corrections or improvements to the manual.

9.3 — 📝 Summarize Page

Click 📝 **Summarize Page** to generate a short summary of the current topic. The summary appears in the chat panel with a colored badge indicating its source. See Section 6.4 for a visual walkthrough and Section 6.5 for copying the result into another app.

10. Troubleshooting

"Chat says 'No direct local topic matches found'"

The AI isn't configured yet. Either:

- Enable local LLM (see Section 8), or
- Save a cloud API key (see Section 7 — we recommend **Groq** for speed)

"Local LLM toggle is unchecked / can't find it"

The toggle defaults to OFF. Open 🖥️ **Local LLM (Offline)** and turn it on after installing Ollama.

"My API keys disappeared after restarting"

Keys are saved to your browser's local storage for `http://127.0.0.1:39820`. They persist across restarts. They do **not** carry over to a different browser, a different install folder, or if you clear browser site data.

> 💾 **Where are my keys stored?** Your API keys and Local LLM settings are saved privately in your browser's Local Storage, not on a cloud server. They are tied specifically to `127.0.0.1:39820`. Uninstalling the app will *not* clear them, ensuring they survive future software updates. To wipe them manually, use the 🗑️ **Clear All Keys** button in the Cloud AI panel.

"Local LLM failed — connection refused"

Ollama isn't running. Start it from the Start menu, or launch `ollama app.exe` manually. Verify with `Invoke-WebRequest http://localhost:11434/api/tags`.

"CORS error when using local LLM"

Ollama blocks browser requests by default for security. If you see a CORS error in the browser console (F12), you need to set an environment variable and restart Ollama.

HOW TO SET OLLAMA_ORIGINS AND RESTART:

1. **Find Ollama:** Look for the Ollama icon (a white llama face) in your Windows **System Tray** (bottom-right corner of your screen, near the clock). You may need to click the ^ arrow to see hidden icons.
2. **Quit Ollama:** Right-click the icon and select **Quit Ollama**. (The server must be completely closed for the new setting to take effect).
3. **Set the variable:**
   - Press `Win` + `R`, type `sysdm.cpl`, and press Enter.
   - Go to the **Advanced** tab → click **Environment Variables**.
   - Under **User variables**, click **New**.

- Variable name: `OLLAMA_ORIGINS`
- Variable value: `http://127.0.0.1:39820`
- Click OK on all windows to save.

4. **Restart Ollama:** Open your Windows Start Menu, search for **Ollama**, and click it to launch the app again. The icon will return to your system tray.
5. Refresh the KnowledgeDesk browser tab.

"The first summary takes 2 minutes"

The local model is loading into RAM. This is a cold start — it happens once per session. Subsequent requests are much faster (5–20 seconds).

❓ Why does my browser say "127.0.0.1 refused to connect"?

This means the KnowledgeDesk server has been closed. If you close the minimized black terminal window in your taskbar, Windows stops the server, and the browser loses its connection.

To fix this: Simply relaunch KnowledgeDesk. You can find it in any of these places:

- Your **Desktop** shortcut (if you created one during setup)
- The **Start Menu** (search for "KnowledgeDesk")
- Your **Pinned apps** in the Start Menu
- The **Recently added** section of the Start Menu

Once the app relaunches, the server will start again in the background and you can refresh the browser page.

11. Getting Help

Report bugs, unclear UX observations, or feature requests by email:

> **Contact:** osfinancials1@gmail.com

All feedback is read. Response time depends on volume.

Developer: © Dan van Zyl - 2026